Shannen Lee

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Portfolio | GitHub | LinkedIn

SUMMARY

Full-stack software engineer with experience across backend systems, frontend interfaces, and deployment infrastructure, with a background in applied machine learning and robotics.

WORK EXPERIENCE

ST Engineering **Sep 2025 – Present**

Software Engineer - Attached to The Digital and Intelligence Service

- Developed a reusable, low-code web application builder in Dash that lets non-technical teams dynamically generate custom forms and data visualisations from JSON/YAML config files, adopted across 6 different use cases without writing new application code.
- Designed configurable Kafka and PostgreSQL integrations so new services can plug into event-driven messaging and persistent storage through configuration alone, cutting integration effort for each new use case from days to hours.
- Automated deployment across multiple OpenShift staging environments using Helm charts and Argo CD GitOps, replacing a manual multi-step setup with a single repeatable process.

DSO National Laboratories **Jul 2024 – Dec 2024**

Machine Learning Engineer Intern

- Designed and implemented a hierarchical reinforcement learning system in PyTorch for coordinating 12 agents in a simulated multi-agent environment.
- Tuned state, observation, and reward configurations for the system, lifting the task success rate above 95%

Continental Automotive Singapore **Jul 2023 – Dec 2023**

Software Engineer Intern

- Led development of a fleet management system for autonomous food delivery robots using FastAPI, ROS2 and OpenRMF, with a real-time dashboard tracking fleet status, including battery, task state, and location.
- Deployed the system to a fleet of 3 robots operating concurrently along a 6-waypoint delivery route, demonstrating live deliveries to key stakeholders.

AMARIS.AI **Jul 2022 – Dec 2022**

Software Developer Intern

- Built reusable, multilingual React components and extended FastAPI backend features for an AI-powered paint-visualisation platform used by customers across 8 APAC countries.
- Resolved 30+ bugs across photo processing, billing, and report downloads, and built a 70+ test backend suite that caught regressions before release, improving platform reliability for production users.

PROJECTS

Where to Touch Grass | View Project

- Built an interactive map and guide to 90+ parks, nature reserves and trails in Singapore, with a page per spot covering entrances, travel time and photos, using Next.js, TypeScript, MapLibre and Supabase.

Health Tracker Bot | View Project

- Built and self-hosted a Telegram bot that estimates a meal's nutrition from a photo and charts progress over time, with 1,000+ meals logged over 2 months; deployed on a home server with Docker behind a Cloudflare Tunnel.

EDUCATION AND CERTIFICATIONS

Nanyang Technological University, Singapore **Aug 2020 – Dec 2024**

Bachelor of Computing in Computer Science with Honours (Distinction)

AWS Certified Machine Learning Engineer – Associate **May 2025**

Amazon Web Services | View verified credential on Credly

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C, C++, Dart, HTML, CSS

Frameworks: React, FastAPI, Flask, Dash, Django, Flutter, ROS2, PyTorch, TensorFlow

Tools: Docker, Kubernetes, OpenShift, Helm, Argo CD, Kafka, PostgreSQL, MariaDB, Firebase, Supabase, MapLibre, Git, Linux, Bash; AWS EC2, S3, Lambda, ECR, Athena, SageMaker